

Cyclobutanedicarboxylate Metal-Organic Frameworks as a Platform for Dramatic Amplification of Pore Partition Effect

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KEYWORDS: *pacs*, multi-module MOF, cyclobutanedicarboxylate, ultrafine pore tuning, bioisosteric

ABSTRACT: Ultrafine tuning of MOF structures at sub-angstrom or picometer levels can help improve separation selectivity for gases with subtle differences. However, for MOFs with large-enough pore size, the effect from ultrafine tuning on sorption can be muted. Here we show an integrative strategy that couples together extreme pore compression with ultrafine pore tuning. This strategy is made possible by unique combination of two features of the *pacs* platform (*pacs* = partitioned *acs*): multi-modular framework and exceptional tolerance towards isorecticular replacement. Specifically, we use one module (Ligand 1, L1) to shrink pore size to an extreme minimum on *pacs*. A compression ratio of about 30% was achieved (based on unit-cell *c/a* ratio) from prototypical 1,4-benzenedicarboxylate-*pacs* to *trans*-1,3-cyclobutanedicarboxylate-*pacs*. This is followed by using another module (Ligand 2, L2) for ultrafine pore tuning (< 3% compression). This L1-L2 strategy increases C₂H₂/CO₂ selectivity from 2.6 to 20.8 and gives rise to excellent experimental breakthrough performance. Being the shortest cyclic dicarboxylate that mimics *para*-benzene-based moieties using bioisosteric (BIS) strategy on *pacs*, *trans*-1,3-cyclobutanedicarboxylate-based MOF chemistry offers new opportunities in MOF chemistry.

Because size matching between pore and gas molecules plays an important role in sorption properties, synthetic design of small-pore MOFs holds promise for optimizing small-gas separation efficiency.¹⁻⁷ Currently, ultrafine pore tuning in small-pore regime using isorecticular chemistry remains a challenge.⁸⁻¹⁰ MOF platforms often consist of just one inorganic module and one organic module, a feature that can limit the scope of ultrafine pore tuning.¹¹⁻¹⁴ One scenario is that an isorecticular change to ligands (in a long-range ordered fashion) would typically impact all framework ligands, causing a pore-size change^{15,16} that can overshoot desired values and consequently miss optimal separation metrics.

For ultrafine pore tuning, a multi-module platform can offer extra opportunities.^{8,17-23} Independent tuning of each module can allow for pore change in smaller increments and in controlled crystallographic directions so that both pore size and shape can be tuned. Furthermore, a multi-module system offers more structural variations, which translates into additional opportunities.

Pore space partition (PSP) is intrinsically suited to produce small-pore multi-modular MOFs.²⁴⁻²⁷ On *pacs* platform, PSP naturally introduces another module (pore-partitioning ligand) to divide up pore space, leading to both multi-modular MOFs and smaller pore. The *pacs* type has general formula of [(M1)_x(M2)_{3-x}(O/OH)(L1)₃(L2)]_n,^{24,28-47} where **L1** is a ditopic ligand for forming the *acs* net (MIL-88/MOF-235),⁴⁸⁻⁵⁰ and **L2** is a pore-partitioning ligand enabling the formation of *pacs*.

When MOFs have sufficiently large pore size, ultrafine tuning, even if accomplished structurally, can have little impact on separation because the change is too subtle to be felt differently by subtly different gas molecules. We reason that if we compress pore size to an extreme minimum for a given platform, the effect of subsequent ultrafine pore tuning can be amplified. Here, we show a strategy that combines extreme pore tightening with ultrafine pore tuning to boost C₂H₂/CO₂ selectivity.

Inspired by bioisosteric (BIS) replacement method, which is commonly practiced in drug design by replacing benzene rings with other scaffolds,⁵¹ we select *trans*-1,3-cyclobutanedicarboxylic acid (**H₂tcb**) as L1 ligand, resulting in the synthesis of a series of *pacs*-MOFs (**CPM-110** family). For ultrafine tuning, three L2 ligands, 2,4,6-tris(4-pyridyl)-1,3,5-triazine (**tpt**), 2,4,6-tris(4-pyridyl)pyridine (**tp_{py}**) and 1,3,5-Tri(pyridin-4-yl)benzene (**tpbz**) are used. CPM-110 can be further expanded by changing trimer types, but cobalt-vanadium (CoV) trimer is the focus here. Members of CPM-110 are denoted **CPM-110a-CoV** (same as **CoV-tcb-tpt**), **CPM-110b-CoV** (**CoV-tcb-tp_{py}**) and **CPM-110c-CoV** (**CoV-tcb-tpbz**) where **a**, **b**, and **c** denote L2 type. We show that changing L2 cause little difference in C₂H₂/CO₂ selectivity (2.6-2.8) when L1 is bdc (Scheme 1). Although bdc is small by MOF standard, it is large enough to conceal ultrafine tuning effect by L2. Revealingly, after replacing bdc with tcb, the change in L2 causes large differences in sorption properties, especially C₂H₂/CO₂ selectivity (12.3-20.8).

Scheme 1. Illustration of pore compression with L1 and ultrafine pore tuning with L2 and their effects on C₂H₂/CO₂ selectivity.

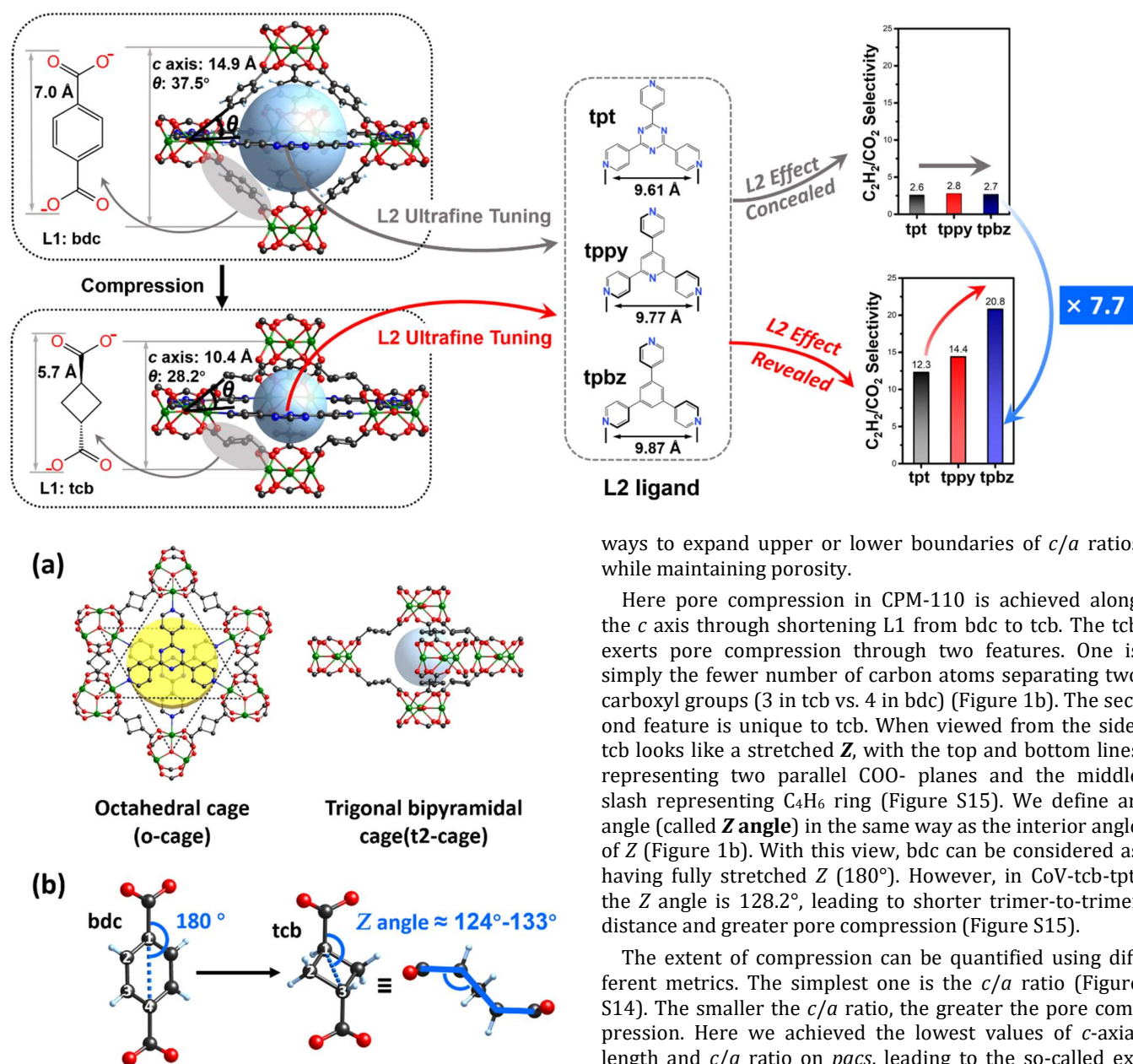


Figure 1. (a) Octahedral and trigonal bipyramidal cages in CPM-110. (b) the definition of Z angle and stretched-Z configuration of tcb ligand.

CPM-110-CoV series crystallized in hexagonal $P6_3/m$ (Tables S1-S3). The *pacs* has two cage types: octahedral cage (**o-cage**) made of six trimers and trigonal bipyramidal cage made of five trimers (**t2-cage**) (Figure 1a). Using different combinations of L1 and L2, these cages can be controlled to be regular polyhedra (at $c/a = 1.15$ for t2-cage or 1.633 for o-cage), elongated or compressed polyhedra (Figure S14). It is both fascinating and important to find

ways to expand upper or lower boundaries of c/a ratios while maintaining porosity.

Here pore compression in CPM-110 is achieved along the c axis through shortening L1 from bdc to tcb. The tcb exerts pore compression through two features. One is simply the fewer number of carbon atoms separating two carboxyl groups (3 in tcb vs. 4 in bdc) (Figure 1b). The second feature is unique to tcb. When viewed from the side, tcb looks like a stretched Z, with the top and bottom lines representing two parallel COO⁻ planes and the middle slash representing C₄H₆ ring (Figure S15). We define an angle (called **Z angle**) in the same way as the interior angle of Z (Figure 1b). With this view, bdc can be considered as having fully stretched Z (180°). However, in CoV-tcb-tpt, the Z angle is 128.2°, leading to shorter trimer-to-trimer distance and greater pore compression (Figure S15).

The extent of compression can be quantified using different metrics. The simplest one is the c/a ratio (Figure S14). The smaller the c/a ratio, the greater the pore compression. Here we achieved the lowest values of c -axial length and c/a ratio on *pacs*, leading to the so-called extreme compression. The a and c lengths correspond to the distances between trimers along each direction (Figure S16). For a given trimer, the a axis is determined by L2, but the c axis is determined by both L1 and L2 (Figure S17). For three L2 ligands here, the a axis length stays very close to each other, however, the c axis length decreases dramatically from bdc (~7.0 Å between O-atoms) to tcb (~5.7 Å), leading to the c axis of 9.83 Å and the c/a ratio of 0.57 in CoV-tcb-tpbz-act (**act = activated**), both of which are the lowest in reported *pacs* (Figure 2, Tables S11-S13). In comparison, in CoV-bdc-tpbz, the a and c axes are 17.09 and 14.61 Å, respectively with the c/a ratio of 0.86.

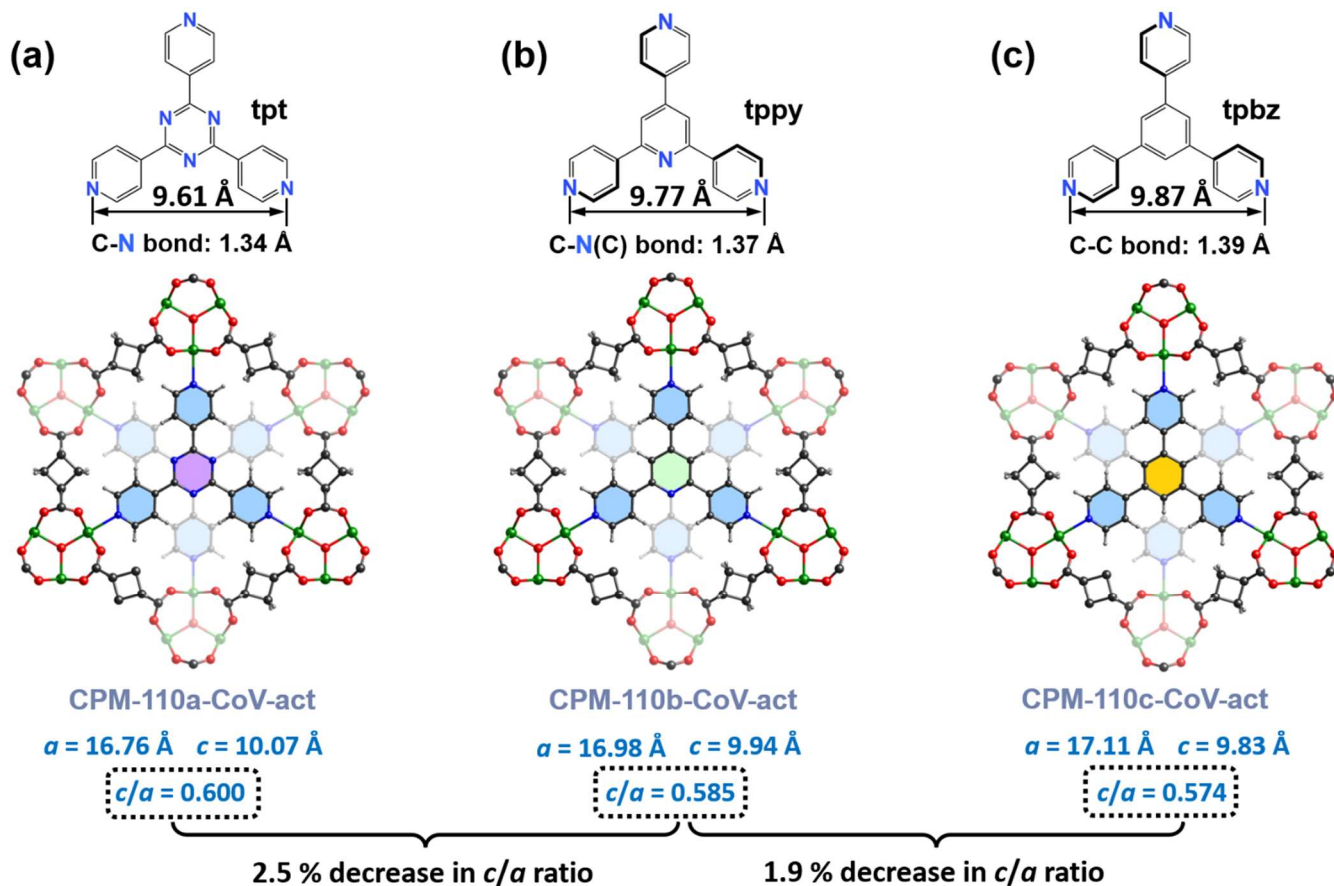


Figure 2. Ultrafine tuning when a subtle change in bond length (about 0.05 Å) propagates to L2 size and c/a ratio.

The compression of t2-cage can be visualized using an angle formed between one edge and the equatorial plane. This angle, called the **flattening angle** (θ angle) here, is 54.7° for two face-sharing regular tetrahedra and becomes 45° for a regular trigonal bipyramid (Figure S14). In CoV-bdc-tpt, it is down to 37.5° (Scheme 1). A further compression with L1 in CoV-tcb-tpt reduces it to just 28.2° (Scheme 1, Figure S18). Ultrafine tuning with L2 can adjust this angle in the narrow range from 28.2° in CoV-tcb-tpt to 26.5° in CoV-tcb-tpbz-act.

The compression of o-cage can be geometrically quantified with octahedron-based metrics (Figure S14). However, there is a chemically meaningful metric: the distance between adjacent tpt planes. In CoV-bdc-tpt, it is 7.5 Å. It is compressed to 5.2 Å in CoV-tcb-tpt, significantly closer to the limit set by non-porous phase (3.40 Å in graphite, Figure S19). The pore volume of CoV-tcb-tpt is 31.6%, smaller than CoV-bdc-tpt (56.1%). These pore metrics illustrate an extreme case of pore compression while still retaining a high degree of porosity in CPM-110.

Ultrafine pore tuning by L2 ligands can be appreciated from the actual dimensions of L2 in CPM-110 series. Because the sample activation is a single-crystal-to-single-crystal process, single-crystal X-ray diffraction was performed to characterize structures of all activated CoV-tcb phases (Tables S1-S3). For the activated structures, the edge lengths of L2 ligands are 9.61, 9.77, and 9.87 Å in

CoV-tcb-tpt-act, CoV-tcb-tpy-act, and CoV-tcb-tpbz-act, respectively, largely due to the difference between aromatic C-N (1.34 Å) and C-C bonds (1.39 Å) (Figure 2). The difference between adjacent members (tpy and tpbz) is as small as 0.10 Å. Despite such tiny differences in ligand size, they generate measurable differences in unit cell dimensions, from 16.76/10.07 Å to 16.98/9.94 Å to 17.11/9.83 Å for the a/c axes in CoV-tcb-tpt-act, CoV-tcb-tpy-act, and CoV-tcb-tpbz-act, respectively (Tables S11-S13). The increase in the a axis and the decrease in the c axis are both caused by the size increase of L2 ligands (Figure 2, Figure S20). Overall, the c/a ratio decreases by 2.5% from tpt to tpy and by 1.9% from tpy to tpbz (Table S13). Such ultrafine tuning by L2 can be contrasted with the extreme compression caused by L1 change where the c/a ratio decreases by 30.0% (from 0.884 for CoV-bdc-tpt to 0.619 for CoV-tcb-tpt).

Gas sorption studies reveal inspiring effects of coupling together extreme pore compression using L1 and ultrafine pore tuning using L2. We note that in the absence of L1 compression, ultrafine tuning with L2 is ineffective. As evidence, CoV-bdc-L2 series (L1 = bdc) show similar N_2 uptake at 77 K (Figure 3a, Figure S21) while CoV-tcb-L2 series (L1 = tcb) show far more significant N_2 uptake difference (Figure 3a).

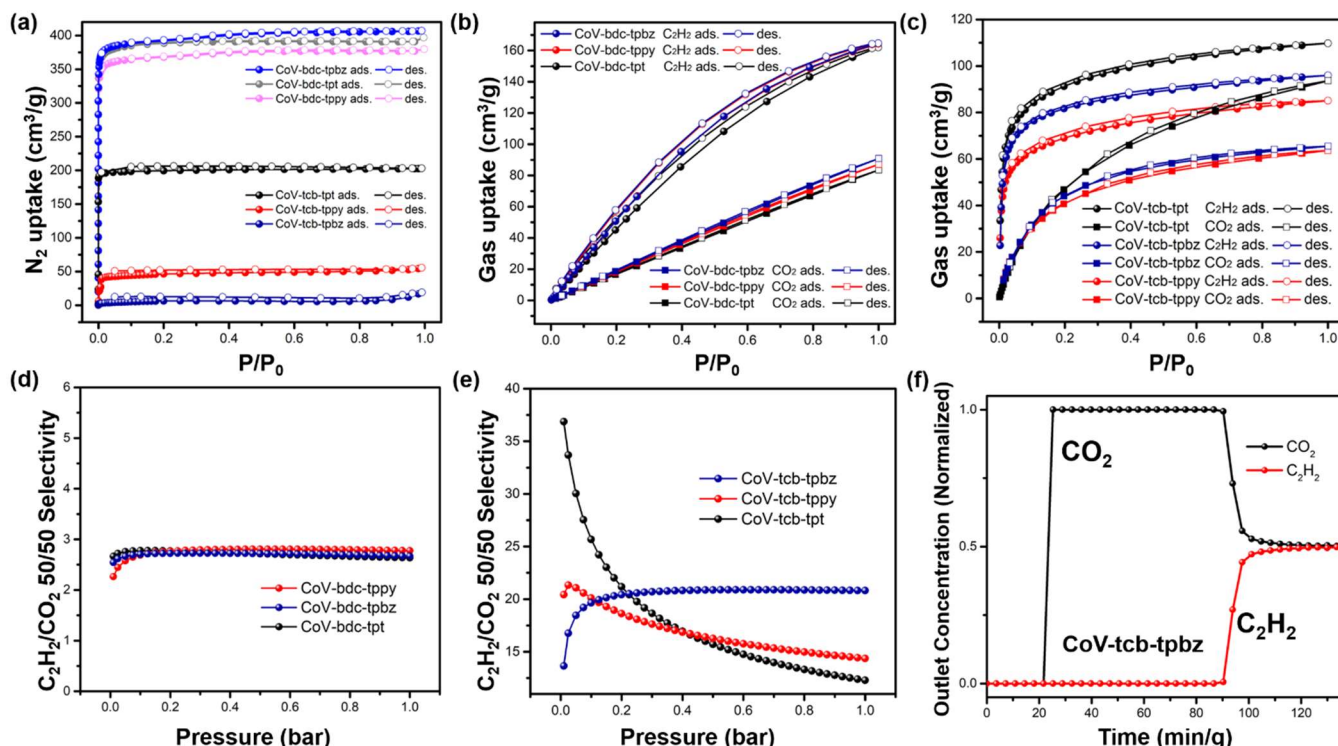


Figure 3. Gas sorption properties of CoV-bdc and CoV-tcb *pacS*. (a) N₂ uptake at 77 K. (b) and (c) C₂H₂ and CO₂ uptake at 298 K for CoV-bdc and CoV-tcb *pacS*, respectively. (d) and (e) C₂H₂/CO₂ (50/50, v/v) selectivity for CoV-bdc and CoV-tcb *pacS*, respectively. (f) The experimental breakthrough curves for equimolar C₂H₂/CO₂ gas mixtures of CoV-tcb-tpbz at 298 K and 1 atm.

Furthermore, the muted or amplified effect of L2 tuning is also evident in C₂H₂/CO₂ sorption. Without L1 compression, CoV-bdc-tpt, CoV-bdc-tpy and CoV-bdc-tpbz perform similarly by adsorbing 83.6, 87.2 and 90.9 cm³/g CO₂ and 162.0, 164.3 and 164.7 cm³/g C₂H₂ at 298 K and 1 atm, respectively (Figure 3b, Figures S26-S27). There is little difference in IAST separation selectivities for C₂H₂/CO₂ (50/50, v/v) which are 2.6, 2.8 and 2.7 for CoV-bdc-tpt, CoV-bdc-tpy and CoV-bdc-tpbz, respectively (Figure 3c). In comparison, with L1 compression, CoV-tcb *pacS* shows pronounced differences in CO₂, C₂H₂ uptake and separation selectivity. CoV-tcb-tpt, CoV-tcb-tpy and CoV-tcb-tpbz adsorb 93.6, 63.7 and 65.5 cm³/g CO₂ and 109.8, 85.0 and 96.0 cm³/g C₂H₂, respectively (Figure 3d, Figures S29-S32). The C₂H₂/CO₂ (50/50, v/v) IAST separation selectivity of CoV-tcb-tpt, CoV-tcb-tpy and CoV-tcb-tpbz are 12.3, 14.4 and 20.8 (Figure 3e), respectively, based on the dual-site Langmuir-Freundlich (DSLFF) fitting (Table S16). C₂H₂/CO₂ selectivity of CoV-tcb-tpbz of 20.8 is the highest among the reported CoV-*pacS* materials (Table S17), and is higher than some well-known MOFs such as CPM-111a-Ni³⁸ (20.1), MIL-160⁵² (10.0), UAST-74a⁵³ (9.0) and FJU-90a⁴⁴ (4.3) (Tables S18).

To further probe the synergy between L1 and L2, another L1 ligand, bicyclo[1.1.1]pentane-1,3-dicarboxylic acid (H₂bcp) whose size (6.0 Å) falls between bdc (7.0 Å) and tcb (5.7 Å), was used to synthesize CoV-bcp-tpt, CoV-bcp-tpy and CoV-bcp-tpbz (Tables S10-S12). Compared to bdc, bcp does help boost C₂H₂/CO₂ selectivity to 10.0, 10.9, 10.9, respectively. But the difference between these values remains small (Figures S28, S48, Table S14). This lends sup-

port to our strategy that a pre-pore-compression step by L1 helps reveal the ultrafine tuning effect by L2.

The uptake of C₂H₂ at low pressure (0.1 atm) and the ratio (0.1 atm/1.0 atm) indicate the affinity for C₂H₂. The C₂H₂ uptake of CoV-tcb-tpbz at 0.1 atm is 76.5 cm³/g (298K), higher than benchmark MOFs, like ZJU-74a¹ (~72 cm³/g) and FeNi-M'MOF⁵⁴ (~74 cm³/g) (Table S19). The uptake ratio of 0.1 atm/1 atm of CoV-tcb-tpbz is 79.7%, higher than FeNi-M'MOF⁵⁴ and NKMOF-1-Ni⁵⁵ of 77.1% and ATC-Cu⁵⁶ of 75.8% (Table S19).

CoV-tcb-tpbz shows great separation performance on the fixed-bed breakthrough measurement with 2 mL/min⁻¹ total gas flow rate (C₂H₂/CO₂ = 1/1, v/v) at 298 K. As shown in Figure 3f, C₂H₂/CO₂ mixture could be separated by CoV-tcb-tpbz. CO₂ quickly broke through from the column, whereas C₂H₂ was retained in the column for additional ~70 min/g. Multicycle mixed-gas breakthrough experiments are carried out under the same conditions. Breakthrough time is the same for three runs of experiment (Figure S49).

In summary, using bioisosteric strategy of mimicking *para*-benzene-based ligands, we have succeeded in introducing *trans*-1,3-cyclobutanedicarboxylate into MOFs. While structurally similar to *para*-benzene-based systems, the pore size and geometry are dramatically reduced. Such extremely compressed pore geometry together with ultrafine tuning with pore-partition ligands leads to dramatic improvement in C₂H₂/CO₂ separation performance.

ASSOCIATED CONTENT

Supporting Information.

Experimental Procedures and compound characterization data (PDF).

CCDC 2255685-2255688, 2255575-2255576, 2266836, 2266838, 2266840, 2283428-2283429, 2283434 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, U.K.; fax: +44 1223 336033.

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ACKNOWLEDGMENT

We acknowledge the support of this work by the US Department of Energy, Office of Basic Energy Sciences, Materials Sciences and Engineering Division under Award No. DE-SC0010596 (P.F.). Single-crystal X-ray diffraction studies were performed on an instrument purchased with an NSF MRI grant (CHEM 2117040, X. B.).

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